

HEMANG NAGAR, M.S.

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STAFF DATA & AI PLATFORM ENGINEER | DISTRIBUTED SYSTEMS | DATA GOVERNANCE | APPLIED AI

PROFESSIONAL SUMMARY

Hands-on data and platform engineering leader with 20+ years designing and building distributed, data-intensive systems across telecom, federal, and enterprise environments. Specializes in turning ambiguous operational, mission, and governance requirements into explicit, testable system rules, scalable data pipelines, and enforceable platform behavior. Deep experience across backend/platform architecture, large-scale data engineering, distributed processing, cloud services, and AI-enabled applications. Current work supporting ODNI and Intelligence Community programs includes an employer-sponsored governed-data-platform proof of concept spanning data quality, policy enforcement, provenance, lineage, standards-based interchange, and governed semantic outputs; prior work includes high-volume telecom event-correlation and production AI/retrieval systems. Python, Java, and Scala remain core implementation tools rather than the center of the engineering role.

CORE COMPETENCIES

SYSTEMS & PLATFORM ARCHITECTURE	Distributed Systems, Data-Intensive Services, Microservices, API Design, Rule-Driven Workflows, Modularity, Reliability, Observability
DATA ENGINEERING & ANALYTICS	Spark, Hadoop, HBase, Phoenix, SQL, Event Correlation, Temporal Matching, Data Quality, Large-Scale Log Analytics, Batch & Stream Processing
DATA GOVERNANCE & INTEROPERABILITY	Policy Enforcement, Provenance, Lineage, Standards-Based Exchange, OpenLineage, DataHub, Knowledge Graphs, PII Controls
APPLIED AI & RETRIEVAL	LLM Applications, RAG, Hybrid Retrieval, Semantic Search, BM25, Vector Search, Grounded Q&A, Evaluation, Voice AI
IMPLEMENTATION TECHNOLOGIES	Python, Java, Scala, FastAPI, REST APIs, AWS, Docker, DuckDB, SQL, Event-Driven Processing

SELECTED ACCOMPLISHMENTS

- Designed and implemented an employer-sponsored governed-data-platform POC that converts approved data profiles into enforceable processing behavior: quality gates, policy and PII controls, provenance, lineage, standards-based emissions, catalog publishing, and governed semantic outputs. Kept new capabilities modular and validated architectural and behavioral claims with 1300+ automated tests.
- Engineered high-volume telecom correlation workflows that join OTP/network-flow blocking data with SMS/A2P events across time windows, canonical application mappings, shortcode mappings, subscriber identities, and explicit deduplication rules. Evolved the logic as real data exposed edge cases such as mock blocks, cross-application OTP traffic, duplicate matches, and conflicting uniqueness metrics.
- Architected and implemented a production RAG platform over telecom sources including OTP blocking logs, OTPDR events, SMS_A2P files, DNS activity, and network-flow telemetry, combining normalization, metadata-aware context assembly, hybrid retrieval, vector search, BM25 ranking, and grounded analyst workflows.
- Developed an AI-powered outbound voice engagement platform using GPT, Whisper STT, Twilio Voice, Azure Cognitive Services, and real-time orchestration, reducing outbound calling labor by 70%.

PROFESSIONAL EXPERIENCE

KYM ADVISORS INC. | February 2022-Present

ENGINEER / TECHNICAL LEAD - ODNI / INTELLIGENCE COMMUNITY PROGRAMS

Part-time: February 2022-May 2026 | Full-time: May 2026-Present

- Serve as a hands-on engineering and architecture lead supporting ODNI and Intelligence Community data interoperability, governed data, AI/ML, and mission-focused platform initiatives; translate ambiguous mission and governance requirements into implementable architectures, explicit system rules, and working prototypes.
- Architect and implement an employer-sponsored governed-data-platform proof of concept that turns approved data profiles into enforceable behavior across data quality, policy and PII controls, provenance, lineage, standards-based interchange, catalog publishing, and governed semantic outputs.
- Designed the GDP around explicit architectural invariants, human approval and authority boundaries, modular capabilities, and testable refusal behavior; integrated structured and standards-based data formats and validated system claims with a 1300+ automated-test suite.

- Led architecture and development of the Information Interoperability Framework (IIF) for ODNI, translating cross-agency interoperability requirements into platform structure, integration patterns, implementation direction, and technical roadmaps.
- Built a requirements-gathering framework that converted multi-agency mission needs into structured technical inputs, improving traceability, documentation fidelity, and solution discovery.
- Facilitated architecture and requirements sessions with ODNI, Intelligence Community, and federal partner teams to resolve ambiguous requirements, integration boundaries, standards questions, and delivery priorities while balancing mission constraints with implementation practicality.

Impact: Improved traceability and interoperability across mission stakeholders while shortening the path from ambiguous requirements to testable technical solutions and working prototypes.

MOBIUS WIRELESS SOLUTION, Ashburn, VA | June 2015-May 2026

HEAD OF PRODUCT DEVELOPMENT / CHIEF ARCHITECT

- Led architecture and hands-on engineering for telecom data and analytics platforms processing high-volume SMS, OTP, signaling, DNS, and network-flow data; translated operational questions into explicit data models, correlation rules, measurable outputs, and production workflows.
- Engineered cross-source OTP/SMS correlation pipelines that modeled blocking windows, subscriber events, canonical applications, shortcode-to-application mappings, trigger/disruption semantics, and match-level evidence; designed stable deduplication identities to prevent inflated counts across repeated processing.
- Built ingestion, normalization, enrichment, aggregation, and persistence workflows in Spark, Python, Java, Scala, and SQL; processed large daily datasets and maintained summary and match-level tables supporting repeatable investigation, trend analysis, and operational reporting.
- Refined analytical rules as production data exposed exceptions: separated simulated block events from real disruptions and triggers, distinguished same-window OTP traffic for other applications, reconciled subscriber-level versus event-level metrics, and hardened database updates for repeatable daily runs.
- Architected and delivered a production RAG platform for natural-language investigation of telecom logs, using metadata-aware context assembly, hybrid lexical/vector retrieval, semantic ranking, and grounded response generation to reduce manual investigation time.
- Built Spark-based analytics and ML pipelines for anomaly detection, traffic classification, feature engineering, model evaluation, threshold tuning, and performance optimization across large-scale telecom datasets.
- Modernized Java monoliths into distributed services and API-driven components, operated Hadoop/Spark/HBase platforms, and troubleshot data, infrastructure, performance, and integration failures across production environments.
- Owned technical strategy and delivery across architecture, product direction, engineering standards, sprint planning, releases, and onshore/offshore teams while remaining directly involved in implementation and debugging.

Impact: Accelerated telecom fraud and OTP investigations, improved repeatability and explainability of cross-source analysis, and turned large operational datasets into evidence-backed engineering and analyst workflows.

ORBITALZ LLC, Vienna, VA | October 2023-Present

AI DEVELOPER / AI SOLUTIONS ARCHITECT (Part-Time)

- Architected and built a conversational voice platform integrating LLM reasoning, speech-to-text, text-to-speech, call control, campaign data, and backend APIs for automated real-time outreach.
- Engineered the end-to-end service flow in Python/FastAPI across speech processing, dialogue state, prompt orchestration, retrieval-based personalization, external APIs, and failure handling.
- Designed contextual response and personalization logic using campaign and voter-history data, with analytics and feedback loops to evaluate engagement and improve conversation quality.
- Built call analytics, interaction scoring, and operational dashboards to measure campaign performance and support iterative tuning.

Impact: Reduced outbound calling labor by 70% while increasing the scale and consistency of automated engagement.

ADDITIONAL EXPERIENCE

AGILEX TECHNOLOGIES, INC. | Jul 2014-Jun 2015

PRINCIPAL CONSULTANT / LEAD DEVELOPER

- Enhanced the VA eHMP SOA platform across distributed VMs by optimizing Solr queries, refactoring legacy services, and improving synchronization features to boost reliability and search performance.

ACCENTURE @ MARRIOTT | Jun 2012-Jun 2014

PRINCIPAL CONSULTANT / TEAM LEAD

- Architected enhancements to marriott.com, including the EConfo redesign and Weather.com API integration, improving customer communications and accelerating reservation-system updates.

VERISIGN INC. | Jan 2012-Jun 2012

SENIOR CONSULTANT

RED HAT CONSULTING | Feb 2011-Jan 2012

SENIOR CONSULTANT

- Enhanced GEICO.com workflows using jBPM and Blaze Rules and led the CHP Alfresco migration, delivering a scalable enterprise document-management system.

FANNIE MAE | Aug 2010-Feb 2011

SENIOR CONSULTANT

- Supported and enhanced the RAMAPI Test Harness and production systems across multiple WebLogic versions, reducing authentication troubleshooting time with automated diagnostic scripts.

COMPUTECH INC. @ FCC | May 2009-Aug 2010

TECHNICAL LEAD / SOFTWARE ARCHITECT

- Architected FCC Auctions 2.0 and integrated CGI's MOMENTUM product using SOA patterns, improving auction workflows and eliminating legacy ColdFusion dependencies.

RISKMETRICS GROUP (ISS) | Nov 2002-Apr 2009

TECHNICAL LEAD / SR. SOFTWARE ENGINEER

EDUCATION

- Master of Computer & Systems Engineering - Rensselaer Polytechnic Institute
- Master of Decision Sciences & Engineering - Rensselaer Polytechnic Institute
- Bachelor of Instrumentation Engineering - University of Mumbai, Mumbai, India